

InternNova Data Analytics Internship

Week 1 Assignment: Python Fundamentals for Data Analytics

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Duration: 1 Week (7 Days) | Total Marks: 100

Python source code for all tasks

Screenshots of program outputs

Brief explanation of each task

Objective

This report builds a strong foundation in Python programming for Data Analytics, covering Python basics, data types, control flow, functions, collections, and file handling, applied through a mini Python project (Student Record Management System).

Task 1: Python Basics

Explanation: Takes the user's Name, College Name, and Branch as input using input(), and displays a formatted welcome message with the entered details.

Python Code:

```
# Task 1: Python Basics
print("Welcome to Python Fundamentals for Data Analytics!")

name = input("Enter your Name: ")
college = input("Enter your College Name: ")
branch = input("Enter your Branch: ")

print("\n----- Student Details -----")
print(f"Name      : {name}")
print(f"College   : {college}")
print(f"Branch    : {branch}")
```

Output Screenshot:

```
Welcome to Python Fundamentals for Data Analytics!
Enter your Name:  Brojo Mohan Dutta
Enter your College Name:  Serampore College
Enter your Branch:  B.Sc(Mathematics)

----- Student Details -----
Name      : Brojo Mohan Dutta
College   : Serampore College
Branch    : B.Sc(Mathematics)
```

Task 2: Variables & Data Types

Explanation: Declares variables of four core data types (int, float, str, bool) and uses type() to print each variable's data type alongside its value.

Python Code:

```
# Task 2: Variables & Data Types
age = 24                      # Integer
cgpa = 7.02                   # Float
student_name = "Brojo"       # String
is_placed = False             # Boolean

print("Variable:", age, "-> Type:", type(age))
print("Variable:", cgpa, "-> Type:", type(cgpa))
print("Variable:", student_name, "-> Type:", type(student_name))
print("Variable:", is_placed, "-> Type:", type(is_placed))
```

Output Screenshot:

```
Variable: 24 -> Type: <class 'int'>
Variable: 7.02 -> Type: <class 'float'>
Variable: Brojo -> Type: <class 'str'>
Variable: False -> Type: <class 'bool'>
```

Task 3: Operators

Explanation: Builds a simple calculator that takes two numbers as input and performs Addition, Subtraction, Multiplication, Division, and Modulus using arithmetic operators.

Python Code:

```
# Task 3: Operators - Calculator
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Addition      :", num1 + num2)
print("Subtraction   :", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division      :", num1 / num2 if num2 != 0 else "Undefined (division by zero)")
print("Modulus       :", num1 % num2 if num2 != 0 else "Undefined (division by zero)")
```

Output Screenshot:

```
Enter first number: 3
Enter second number: 5
Addition      : 8.0
Subtraction   : -2.0
Multiplication : 15.0
Division      : 0.6
Modulus       : 3.0
```

Task 4: Conditional Statements

Explanation: Takes marks as input and uses if / elif / else to assign a grade: 90+ -> A, 75-89 -> B, 60-74 -> C, below 60 -> Fail.

Python Code:

```
# Task 4: Conditional Statements - Grade Calculator
marks = float(input("Enter your marks: "))

if marks >= 90:
    grade = "A"
elif marks >= 75:
    grade = "B"
elif marks >= 60:
    grade = "C"
else:
    grade = "Fail"
```

```
print(f"Grade: {grade}")
```

Output Screenshot:

```
Enter your marks: 71
Grade: C
```

Task 5: Loops

Explanation: Demonstrates three loop patterns: a for loop printing numbers 1 to 20, a for loop generating the multiplication table of a user-entered number, and a while loop printing even numbers from 1 to 50.

Python Code:

```
# Task 5: Loops

# 1. Print numbers from 1 to 20 using a for loop
print("Numbers from 1 to 20:")
for i in range(1, 21):
    print(i, end=" ")
print()

# 2. Multiplication table of a number
num = int(input("\nEnter a number for multiplication table: "))
print(f"Multiplication table of {num}:")
for i in range(1, 11):
    print(f"{num} x {i} = {num*i}")

# 3. Print even numbers from 1 to 50 using a while loop
print("\nEven numbers from 1 to 50:")
n = 1
while n <= 50:
    if n % 2 == 0:
        print(n, end=" ")
    n += 1
print()
```

Output Screenshot:

```
Numbers from 1 to 20:  
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
```

```
Enter a number for multiplication table: 9  
Multiplication table of 9:  
9 x 1 = 9  
9 x 2 = 18  
9 x 3 = 27  
9 x 4 = 36  
9 x 5 = 45  
9 x 6 = 54  
9 x 7 = 63  
9 x 8 = 72  
9 x 9 = 81  
9 x 10 = 90
```

```
Even numbers from 1 to 50:  
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
```

Task 6: Functions

Explanation: Defines two user-defined functions - square(n) to calculate the square of a number, and average(a, b, c) to calculate the average of three numbers - and calls both with user-provided input.

Python Code:

```
# Task 6: Functions  
  
def square(n):  
    """Returns the square of a number"""  
    return n * n  
  
def average(a, b, c):  
    """Returns the average of three numbers"""  
    return (a + b + c) / 3  
  
num = float(input("Enter a number to find its square: "))  
print(f"Square of {num} = {square(num)}")  
  
a = float(input("Enter first number: "))  
b = float(input("Enter second number: "))  
c = float(input("Enter third number: "))  
print(f"Average of {a}, {b}, {c} = {average(a, b, c)}")
```

Output Screenshot:

```
Enter a number to find its square: 6  
Square of 6.0 = 36.0  
Enter first number: 7  
Enter second number: 4  
Enter third number: 2  
Average of 7.0, 4.0, 2.0 = 4.333333333333333
```

Task 7: Strings & Collections

Explanation: Demonstrates string methods (upper(), lower(), replace(), find()), list operations (append(), remove(), sort()), tuple creation and indexing, a dictionary storing student information, and set operations (add(), remove()).

Python Code:

```
# Task 7: Strings & Collections

# --- String Operations ---
text = "Data Analytics"
print("Original String:", text)
print("Upper Case:", text.upper())
print("Lower Case:", text.lower())
print("Replace 'Data' with 'Big Data':", text.replace("Data", "Big Data"))
print("Index of 'Analytics':", text.find("Analytics"))

# --- List Operations ---
tools = ["Excel", "SQL", "Python"]
tools.append("Tableau")
print("\nTools after add:", tools)
tools.remove("Excel")
print("Tools after remove:", tools)
tools.sort()
print("Sorted list:", tools)

# --- Tuple Creation and Indexing ---
coordinates = (10, 20, 30)
print("\nTuple:", coordinates)
print("First element:", coordinates[0])
print("Last element:", coordinates[-1])

# --- Dictionary storing student information ---
student = {
    "name": "Brojo Mohan Dutta",
    "branch": "B.Sc(Mathematics)",
    "cgpa": 7.02
}
print("\nStudent Name:", student["name"])
print("Student Dictionary:", student)

# --- Set Operations ---
skills = {"Python", "SQL", "Excel"}
skills.add("Power BI")
print("\nSkills after add:", skills)
skills.remove("Excel")
print("Skills after remove:", skills)
```

Output Screenshot:

```
Original String: Data Analytics
Upper Case: DATA ANALYTICS
Lower Case: data analytics
Replace 'Data' with 'Big Data': Big Data Analytics
Index of 'Analytics': 5
```

```
Tools after add: ['Excel', 'SQL', 'Python', 'Tableau']
Tools after remove: ['SQL', 'Python', 'Tableau']
Sorted list: ['Python', 'SQL', 'Tableau']
```

```
Tuple: (10, 20, 30)
First element: 10
Last element: 30
```

```
Student Name: Brojo Mohan Dutta
Student Dictionary: {'name': 'Brojo Mohan Dutta', 'branch': 'B.Sc(Mathematics)', 'cgpa': 7.02}
```

```
Skills after add: {'Excel', 'Power BI', 'Python', 'SQL'}
Skills after remove: {'Power BI', 'Python', 'SQL'}
```

Task 8: Basic File Handling

Explanation: Creates a text file, writes a short introduction into it using write(), then reopens the file and reads/displays its contents using read().

Python Code:

```
# Task 8: Basic File Handling

# Create and write to a text file
with open("introduction.txt", "w") as file:
    file.write("Hi, my name is Brojo.\n")
    file.write("I am pursuing a career as a Data Analyst.\n")
    file.write("I am currently doing my Data Analytics Internship at InternNova.\n")

# Read and display the file contents
with open("introduction.txt", "r") as file:
    content = file.read()
    print("----- File Contents -----")
    print(content)
```

Output Screenshot:

```
----- File Contents -----
Hi, my name is Brojo.
I am pursuing my career as a Data Analyst.
I am currently doing my Data Analytics Internship at InternNova.
```

Task 9: Mini Python Project: Student Record Management System

Explanation: A menu-driven program that stores student records (name, roll number, branch) in a list of dictionaries. Supports adding a student, displaying all records, searching a student by name, and deleting a record, looping until the user chooses to exit.

Python Code:

```
# Task 9: Mini Python Project - Student Record Management System

students = []

def add_student():
    name = input("Enter student name: ")
    age = input("Enter age: ")
    branch = input("Enter branch: ")
    students.append({"name": name, "age": age, "branch": branch})
    print(f"Student '{name}' added successfully.\n")

def display_students():
    if not students:
        print("No student records found.\n")
        return
    print("----- Student Records -----")
    for s in students:
        print(f"Name: {s['name']} | Age: {s['age']} | Branch: {s['branch']}")
    print()

def search_student():
    name = input("Enter name to search: ")
    found = [s for s in students if s["name"].lower() == name.lower()]
    if found:
        for s in found:
            print(f"Found -> Name: {s['name']}, Age: {s['age']}, Branch: {s['branch']}\n")
    else:
        print("Student not found.\n")

def delete_student():
    name = input("Enter name to delete: ")
    global students
    before = len(students)
    students = [s for s in students if s["name"].lower() != name.lower()]
    if len(students) < before:
        print(f"Student '{name}' deleted successfully.\n")
    else:
        print("Student not found.\n")

def menu():
    while True:
        print("===== Student Record Management System =====")
        print("1. Add Student")
        print("2. Display All Students")
        print("3. Search Student")
        print("4. Delete Student")
        print("5. Exit")
        choice = input("Enter your choice (1-5): ")

        if choice == "1":
            add_student()
        elif choice == "2":
            display_students()
        elif choice == "3":
            search_student()
        elif choice == "4":
            delete_student()
        elif choice == "5":
```

```
        print("Exiting program. Goodbye!")
        break
    else:
        print("Invalid choice. Try again.\n")

if __name__ == "__main__":
    menu()
```

Output Screenshot:

```
===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice (1-5): 1
Enter student name: Brojo Mohan Dutta
Enter age: 24
Enter branch: B.Sc(Mathematics)
Student 'Brojo Mohan Dutta' added successfully.

===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice (1-5): 1
Enter student name: Jaspreet Kaur
Enter age: 25
Enter branch: CSE
Student 'Jaspreet Kaur' added successfully.

===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice (1-5): 1
Enter student name: Rishita Mitra
Enter age: 22
Enter branch: ECE
Student 'Rishita Mitra' added successfully.
```

```
===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice (1-5): 2
----- Student Records -----
Name: Brojo Mohan Dutta | Age: 24 | Branch: B.Sc(Mathematics)
Name: Jaspreet Kaur | Age: 25 | Branch: CSE
Name: Rishita Mitra | Age: 22 | Branch: ECE
```

```
===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice (1-5): 4
Enter name to delete: Jaspreet Kaur
Student 'Jaspreet Kaur' deleted successfully.
```

```
===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice (1-5): 2
----- Student Records -----
Name: Brojo Mohan Dutta | Age: 24 | Branch: B.Sc(Mathematics)
Name: Rishita Mitra | Age: 22 | Branch: ECE
```

```
===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student
4. Delete Student
5. Exit
Enter your choice (1-5): 5
Exiting program. Goodbye!
```